

ASCO: Agentic Settlement & Compliance Orchestrator - Project Overview

1. Project Vision

The Agentic Settlement and Compliance Orchestrator (ASCO) is the core intelligence engine for the UbuntuRemit platform. It is designed to modernize cross-border financial transactions by replacing static, rule-based routing with dynamic, multi-agent negotiation. The primary objective is to execute settlements with maximum capital efficiency while guaranteeing zero regulatory violations under South African Reserve Bank (SARB) and international AML guidelines.

2. The Multi-Agent Architecture

ASCO departs from traditional monolithic AI approaches by employing distinct, persona-driven AI agents that evaluate transactions concurrently. This distributed reasoning model mirrors the operations of a human treasury department.

Agent Persona	Primary Objective	Model Requirement
The Compliance Sentinel	Enforce AML, CFT, and SARB regulatory compliance. Acts as the system "brake."	High-parameter, high-reasoning LLM (e.g., Llama-3-70B) focused on unstructured legal text and risk profiling.

Agent Persona	Primary Objective	Model Requirement
The Liquidity Strategist	Identify the fastest and cheapest settlement rails (e.g., Ripple, SWIFT, PAPSS). Acts as the system "engine."	Math and logic-optimized model capable of rapid quantitative analysis of market spreads and fees.
The Master Orchestrator	Manage state, handle the handshake between agents, and format the final ISO 20022 output.	A deterministic framework (like AutoGen or LangGraph) integrated with strict XML generation rules.

3. Technical Stack

- **Infrastructure:** AMD AI Developer Cloud instances powered by AMD Instinct MI300X accelerators.
- **Software Layer:** ROCm 7.0 for deep hardware optimization, combined with vLLM for high-throughput model serving.
- **Application Layer:** Domain-Driven Design (DDD) implemented via Python/FastAPI to ensure decoupled, scalable microservices.

4. Business Value for Enterprise Implementations

For major South African financial institutions and tech consultancies, ASCO provides three critical advantages:

1. **Sovereign Deployment:** Complete data residency is maintained by avoiding third-party LLM APIs.
2. **Auditability:** The multi-agent debate creates a structured "Thought-Action-Observation"

log, providing auditors with transparent reasoning behind every routed payment.

3. **Resilience:** The system automatically resolves transaction friction through intelligent negotiation rather than rejecting transactions outright.